

1 r.traveltime - A GRASS GIS addon to study overland 2 flow dynamics based on landscape connectivity

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Software

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6 Summary

7 r.traveltime is an addon raster module for the GRASS Geographic Information System (GIS)
8 ([GRASS Development Team et al., 2026](#)). Based on terrain analyses, it computes a map of the
9 travel time required for overland flow to reach the catchment's outlet from any given location
10 within the catchment. This approach uses simplified hydrological computations ([Kilgore, 1997](#);
11 [Melesse & Graham, 2004](#); [Muzik, 1996](#)) and only requires terrain elevation maps as a minimum
12 input.

13 [Figure 1](#) illustrates how this works: The map of terrain elevation ([Figure 1\(a\)](#)) is first analyzed
14 in terms of flow direction (not shown) and flow accumulation, which reflects the size of the
15 upstream catchment area ([Figure 1\(b\)](#)). r.traveltime is designed to compute the travel
16 time of overland flow on hillslope elements and in channel segments - a flow accumulation
17 threshold helps to distinguish between both types of flow. This way, it links flow direction, flow
18 accumulation and surface roughness, which are viewed as the main factors governing landscape
19 connectivity, and computes a map of overland flow travel time ([Figure 1\(c\)](#)). r.traveltime
20 supports three representations of overland flow: (i) Each hillslope cell is considered individually
21 in terms of flow length (shown in [Figure 1\(c\)](#)), (ii) flow length is estimated by identifying
22 the minimum upstream length to the ridge (watershed), or (iii) the maximum flow path is
23 considered. The options are compared using travel-time histograms ([Figure 1\(d\)](#)). These
24 alternative formulations allow the sensitivity of travel-time estimates to the representation of
25 upstream flow length to be explored. An independent SAGA GIS calculation using a different
26 flow-velocity formulation is also shown ("Isochrones variable speed").

27 The histogram representation of the travel time map represents a space-time transformation,
28 which is why approaches like r.traveltime fill a gap between methods that provide single
29 measures of flow time (e.g., longest flow time in a catchment) on one side and more complex
30 fully coupled hydrological-hydrodynamic simulations like r.sim.waterin GRASS GIS on the
31 other. r.traveltime was first announced in 2007 and has since been continuously maintained
32 and adapted to new GRASS GIS versions along with improvements following users' feedback
33 over almost 2 decades.

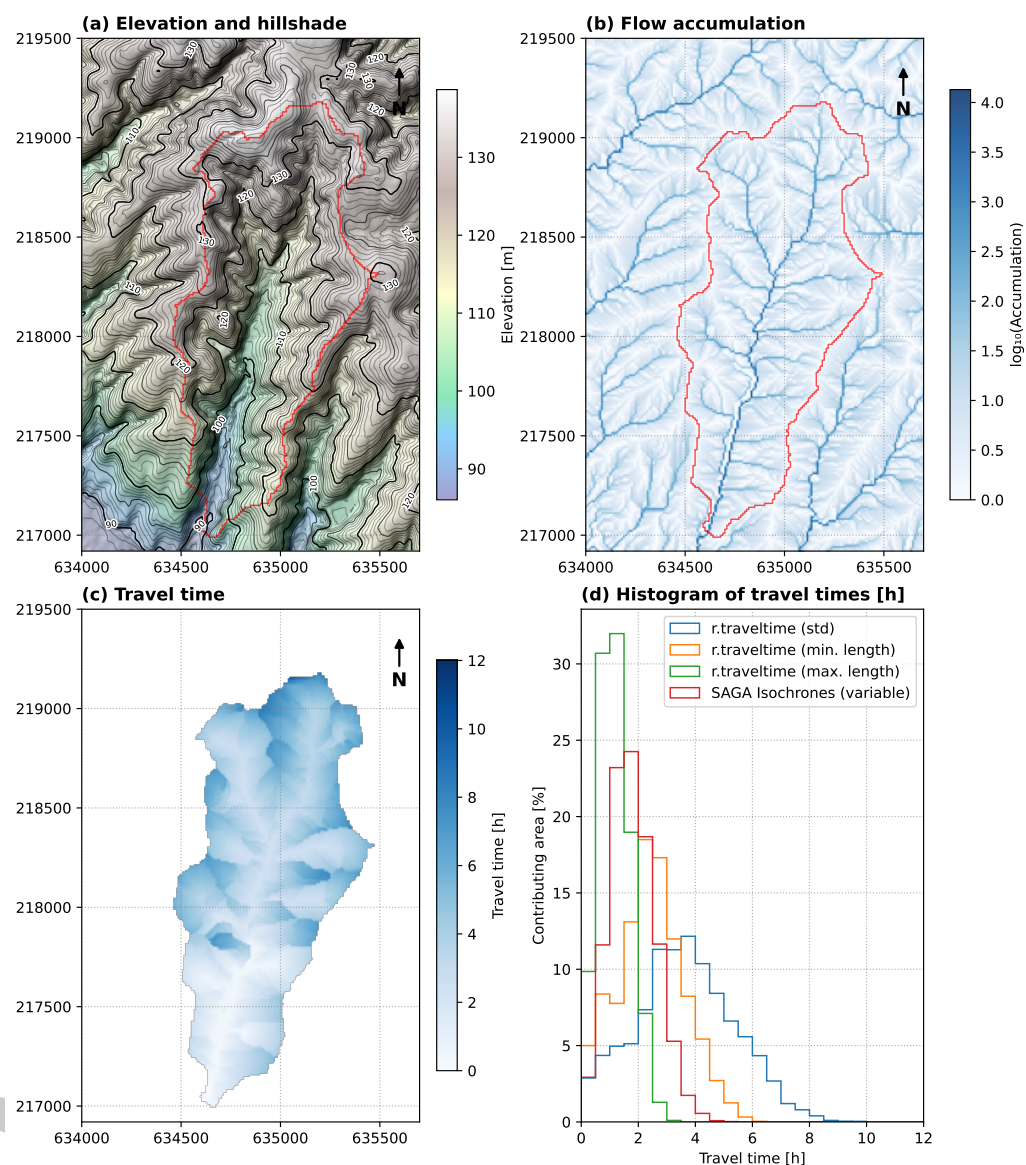


Figure 1: Steps to compute travel times for the North Carolina sample dataset in `r.traveltime`. The Jupyter notebook published alongside the paper includes the code to reproduce these results from downloading the data, installing `r.traveltime` to histogram plotting.

Statement of need

Predicting flooding usually considers a temporal dimension, i.e., it is important to predict when flooding occurs as a response to rainfall. Likewise, hydrological modelling might consider a spatially distributed representation of processes (Refsgaard, 1996), requesting a linkage to GIS data. According to this early reference on hydrological modelling, several types of models exist, whereby the classification is considered along three dimensions: (i) level of processes representation (i.e., complexity), (ii) time dimension (stationary model vs. models with increments of time, e.g., seconds to days), (iii) level of spatial distribution (i.e. spatial aggregation of landscape elements). If flooding is considered, short intervals in time are needed, which covers ranges from sub-hourly to daily time steps. While complex simulation tools exist (Mitasova et al., 2004), which entail a high degree of complexity in all three categories

(physically-based description, small temporal and spatial increments), simpler models still require a parameterization of temporal dynamics, e.g., through provision of measures of flow time in sub-catchments. `r.traveltime` addresses this requirement whenever complex simulations are out of scope for practitioners who need travel time estimates to parameterize their own models. In this specific case, the histogram is used as a unit hydrograph derived from landscape connectivity. An intuitive illustration of this approach and how it could be communicated is provided by Schulz et al. (2018). This approach is in line with the space-time transformation shown in the Figure 1 and goes beyond single measures of maximum flow time in a catchment (Maidment, 1993).

State of the field

GIS enabled terrain-based travel time estimates, often referred to as geomorphological unit hydrographs (Kilgore, 1997; Melesse & Graham, 2004; Muzik, 1996). Approaches similar to `r.traveltime`, besides its broad theoretical foundation from literature, are still rare and either emerge into simplifications in representing flow length and slopes or consider more complex simulations with time steps. Apart from `r.traveltime` in GRASS GIS, there is a tool called “Isochrones variable speed” (Al-Smadi, 1998) in SAGA GIS, which evolved in parallel to and independently from `r.traveltime`. Likewise, Diakakis (2011) describes a multi-step approach to achieve similar results in ArcGIS. Despite their similarity, the approaches differ in their hydrological assumptions (Table 1), highlighting `r.traveltime`’s position between simple and complex methods.

Table 1: Overview of different methods to study flooding under consideration of the Refsgaard (1996) criteria.

Approach	Processes	Spatial distr.	Temporal distr.	Main result	Reference
Flow length	No	Limited	No	Single Flow length to stream	-
Concentration time	Single empirical eq.	Limited	Limited	Single time measure	(Maidment, 1993)
Scalgo Live	Fill sinks	Yes	No	Flooded volume map	https://scalgo.com/
<code>r.traveltime</code>	Simple hydrology	Yes	Yes (indirect)	Travel time map	-
SAGA Isochrones	Simple hydrology	Yes	Yes (indirect)	Travel time map	(Al-Smadi, 1998)
<code>r.sim.water</code>	diffusive wave	Yes	Yes (full)	Modelled water level and velocity map	(Mitasova et al., 2004)
Scalgo TUFLOW	Full Shallow water eq.	Yes	Yes (full)	Modelled water level and velocity map	(Huxley & Syme, 2016)

`r.traveltime` complements rather than replaces these approaches by providing an intermediate level of spatial and temporal representation: it resolves the distribution of travel times across a catchment without requiring a full hydrodynamic simulation. Moreover, through `g.extension` it is a readily available addon to GRASS and can be modified by users according to their own needs.

Software design

`r.traveltime` emerged from a practical need to estimate the distribution of travel times in a catchment. This quantity depends on both upstream (flow accumulation) and downstream information (i.e., the outlet as reference point above which travel times are added along the flow path). This type of calculation is not possible using standard map calculation features, which is why the author decided in 2007 to create a new GRASS GIS raster addon, because of its well documented API and the existence of `r.example`¹, a simple minimum code to develop own addons. A simple realization has been considered through the implementation of a recursive function `ttime()`, which calculates the travel time of water against gravity (i.e. in opposite direction a water drop would travel) from the outlet until it reaches all points along the watershed boundary. A recursive implementation - as summarized in the pseudocode below - acknowledges the cumulative nature of travel time along the upstream flow path, whilst being dependent on upstream area. This formulation avoids solving the problem through a series of steps of individual raster calculations provided by different modules.

```
FUNCTION ttime(ftime):
  FOR each D8 direction:
    IF inflow():
      ttime(ftime+traveltime())
```

Here, `inflow()` checks for tributary cells. For each of these cells, `ttime()` is called, remembering the travel time from the current cell to the outlet `ftime`, which is added to the local travel time between the current and tributary cell being considered.

`r.traveltime` was first announced in 2007 (Förster, 2026) and entered the official GRASS GIS Addons repository in 2012². It has since been maintained across GRASS GIS versions. The author suggested some minor modification to the GRASS GIS developers in 2014 and 2015, which have been committed to the svn repository in 2016, including technical modifications to raster input/output by the GRASS GIS maintainers. `r.traveltime` was also ported to the new GitHub repository in 2019 by the GRASS GIS developers, where it is still maintained and made available³. Users' feedback has been collected over the course of this development timeline and known issues have been compiled in the documentation published alongside the source file in the repositories. The version described in the paper is available in a fork of the official repository: <https://github.com/kristianfoerster/grass-addons/tree/grass8/src/raster/r.traveltime>

Research impact statement

`r.traveltime` has been used in at least seven scientific publications representing six countries (see Table 2). It is worth noting that it has been also applied in contexts other than the original aim of predicting floods, e.g., tool development (Minelli et al., 2010), delineation of water protection areas (Kaboré et al., 2022; Koffi Konan et al., 2014), sediment yield (Berten et al., 2021), and spatial planning (Legarda Garzón, 2020).

Mathematics

The mathematical description, at least for the standard calculation method ("std" in Figure 1), is available in the historical technical description (Förster, 2026). The minimum and maximum approaches include a flow length calculation before calling `ttime()` and the overland flow calculation is replaced by the approach described by Kilgore (1997).

¹<https://github.com/OSGeo/grass/tree/main/doc/examples/raster/r.example> (09 Sep 2026)

²<https://trac.osgeo.org/grass/browser/grass-addons/grass6/raster/r.traveltime> (09 Sep 2026)

³<https://github.com/OSGeo/grass-addons/tree/grass8/src/raster/r.traveltime/> (09 Sep 2026)

Table 2: Selected scientific applications of `r.traveltime`

Reference	Location	Objectives
Suprit et al. (2010)	Talpona River, India	Provision of travel times for flood risk management
Minelli et al. (2010)	3 catchments in Italy	Development of <code>r.clarke.py</code> , which calls <code>r.traveltime</code> to compute hydrographs based on rainfall input; comparison with empirical equation.
Koffi Konan et al. (2014), Koffi et al. (2015)	Lagune Aghien, Côte d'Ivoire	Delineation of protection areas on the lagoon area depending on travel time
Corripio & López-Moreno (2017)	Ésera River catchment, Spain	Computation of travel times in sub-catchment for hydrological modelling
Legarda Garzón (2020)	Abstracted catchments, data from UK	Histogram of travel times utilized to identify locations for natural flood management measures
Berteni (2019), Berteni et al. (2021)	Guerna catchment, Italy	Derivation of a unit hydrograph for rainfall runoff and sediment yield prediction
Kaboré et al. (2022)	Barrages de Ouagadougou, Burkina Faso	Map of travel time employed to identify travel times to three reservoirs and intersection with pollution sources

AI usage disclosure

Generative AI tools were used to assist with setting up the Binder environment, preparing search queries in languages other than English, and improving the language of this manuscript. They were not used to develop the software. All AI-assisted content was reviewed and verified by the author.

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